Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience owning design, implementation, and operation of large-scale distributed AWS infrastructure. Backend automation, high-throughput data pipelines, and fault-tolerant system design using Python, Go, and Java; current research applies AI/ML to systems automation.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, Go, C/C++, JavaScript, TypeScript, SQL.

Distributed Systems & Cloud: AWS (EC2, EFS, IAM, CloudWatch), Kubernetes, ArgoCD, Helm, Terraform, Ansible, Linux, Docker.

Data & Backend: PySpark, Delta Lake, PostgreSQL, MySQL, Redis, REST APIs, FastAPI, Flask.

Developer Tools: Git, GitLab CI/CD, Jenkins, SonarQube, Bash, Vault, Azure, Google Cloud Platform.

AI & Data: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity, running full requirement-to-evaluation SDLC cycle in Python.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Owned design and operation of **Teamcenter X** distributed infrastructure workflows, automating with Python (Boto3) and reducing deployment time **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and fleet-wide services with fault tolerance in mind.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** with reliable rollback and recovery.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault to modernize a large-scale distributed platform.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding and code reviews.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built distributed data pipeline ingesting and processing a **2.3M-row** hourly telemetry dataset across **263 zones** using PySpark and Delta Lake; designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).